PORTOFOLIO

Masdika Aliman

Robotics & Automation Engineer | AI, IoT & Computer Vision



ABOUT ME

Robotics Engineering student and dedicated Robotics & AI Enthusiast with a strong foundation in Robotics Algorithms, Embedded Systems, Automation, and Artificial Intelligence.

As an active member of the Middle-Sized League (MSL) Soccer Robot Team at Barelang Robotics and Artificial Intelligence Laboratory (BRAIL), I have practical experience in developing advanced systems for localization, object detection, kinematics, control systems, and navigation.

Today I apply that foundation as an AI & IoT Developer intern at PT Sat Nusapersada Tbk in Batam — 19 projects, 4 publications, 9 honors.

EDUCATION

State Polytechnic of Batam

Mechatronics, Robotics & Automation Engineering | 2022 — 2026

Bachelor of Engineering (D-IV) · GPA 3.93 · BRAIL Laboratory

SMA Negeri 8 Batam

Senior High School | Batam









WORK EXPERIENCE

PT Sat Nusapersada Tbk

AI & IoT Developer Intern (Jan 2026 — Present)

Industrial AI, computer vision and IoT for manufacturing — PLC integration via Modbus & MQTT, real-time dashboards, AI on NVIDIA Jetson edge devices.

Freelance

Software Programmer (Jun 2025 — Present)

Machine learning, embedded systems, robotics and application development — from real-hardware datasets to monitoring and prediction interfaces.

WORK EXPERIENCE

Center for Robotics and AI, Polibatam

Robotics Engineer (Dec 2022 — Nov 2025)

Localization, navigation, kinematics, strategy and AI object detection for the MSL soccer robot program at BRAIL.

Robot Epson Training

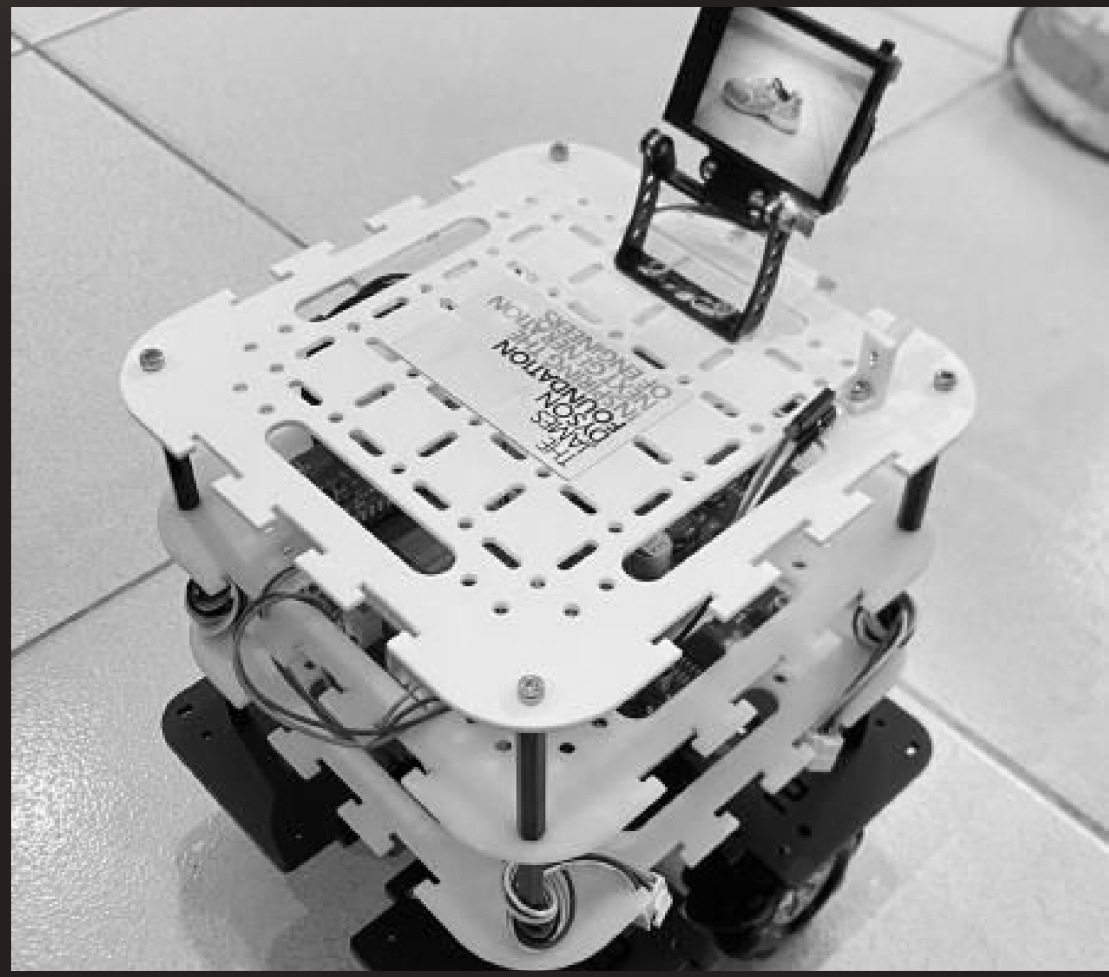
Training Assistant (Sep — Oct 2025)

EPSON RC+ programming, wiring & I/O, setup, calibration and troubleshooting.

Akademi Wirausaha Mahasiswa Merdeka

Embedded Systems Programmer (Sep — Dec 2024)

Trainer Kit Production Division — manuals, learning materials, microcontroller & IoT development.



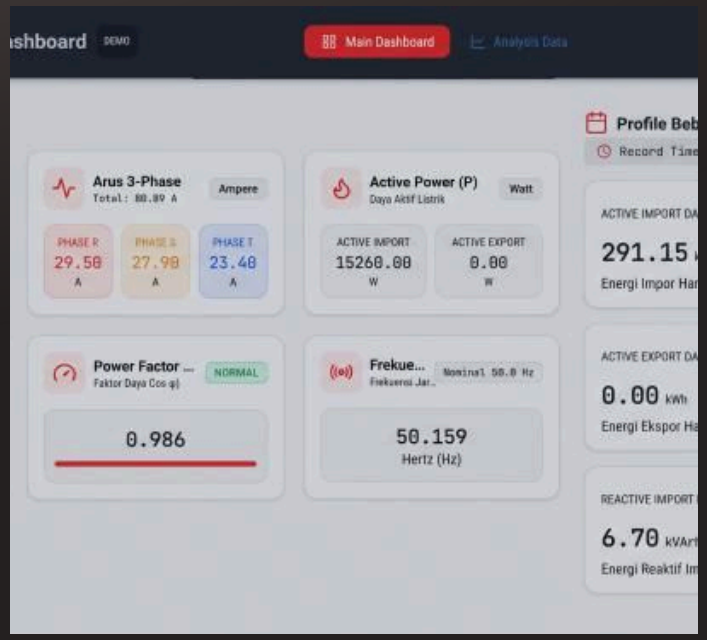
MY WORK

This is the result of my work across industrial AI & IoT, robotics, machine learning, and embedded systems.



Compressor Digital Twin & IoT Telemetry

3D digital twin of an Atlas Copco GA90 compressor — Modbus & HTTP-CGI telemetry, FastAPI, PostgreSQL, digital maintenance logs. PT Sat Nusapersada · Oct 2026



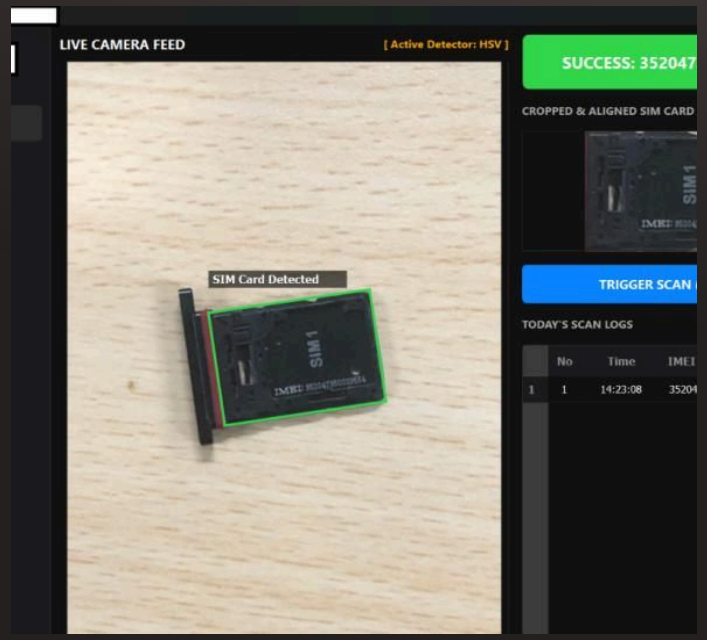
IoT 3-Phase Energy & Power Quality Monitoring

kWh meters over Modbus RTU/TCP, MQTT edge-to-cloud, FastAPI + WebSocket live gauges, trends, and one-click Excel export. PT Sat Nusapersada · Jul 2026



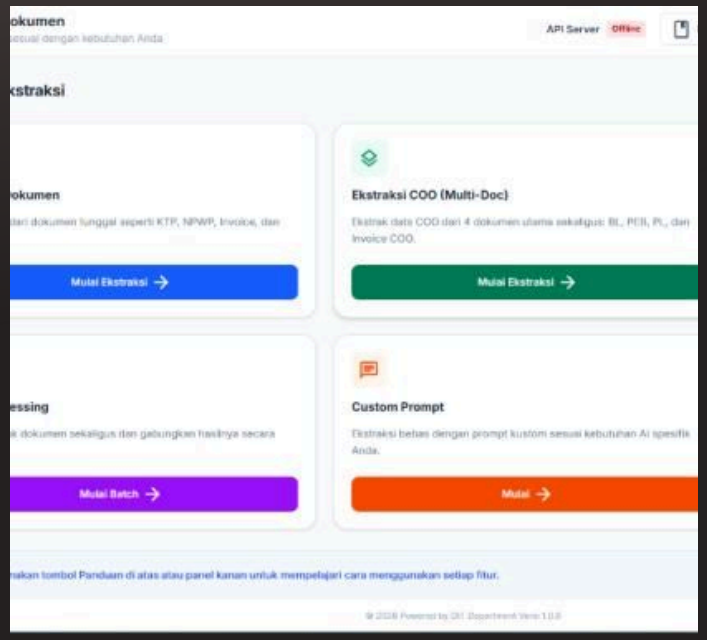
Hygrometer Monitoring & Audit Reporting

Raspberry Pi edge gateway polling hygrometers over serial, MQTT publishing, admin dashboard, automated Cpk compliance reports. PT Sat Nusapersada · Apr–May 2026



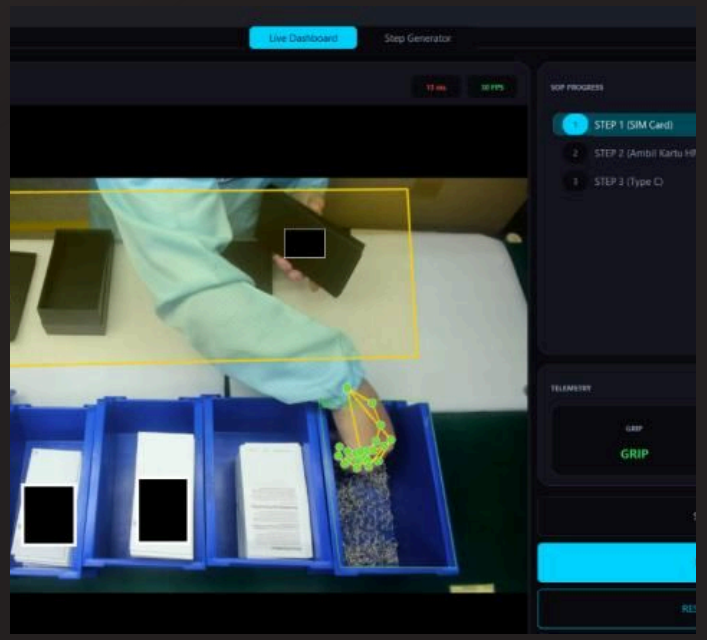
AI-AOI-SIM-CARD

Automated optical inspection with YOLOv12 on industrial cameras, OCR of serialized IMEI, and RPA production logging. PT Sat Nusapersada · Jun 2026



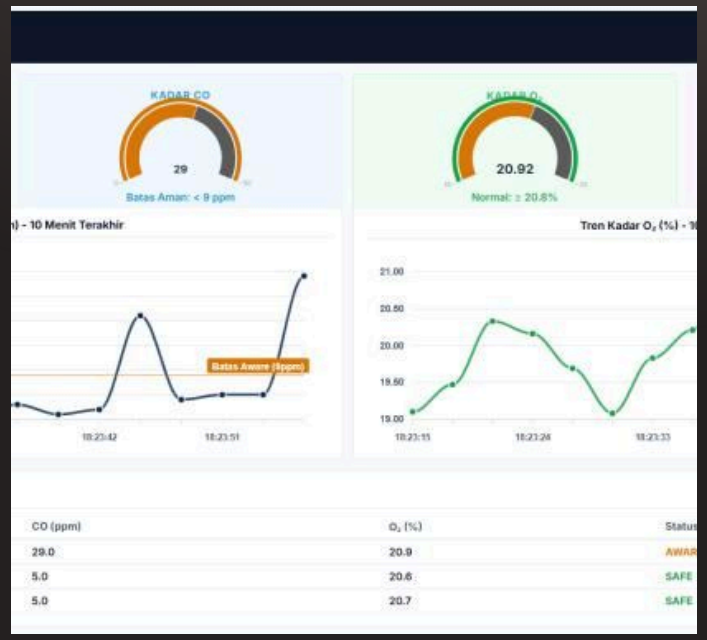
AI Document Extractor

OCR vs multimodal vision-language models benchmarked with CER; PaddleOCR + Qwen-VL pipeline exporting JSON, Excel & Markdown. PT Sat Nusapersada · Mar 2026



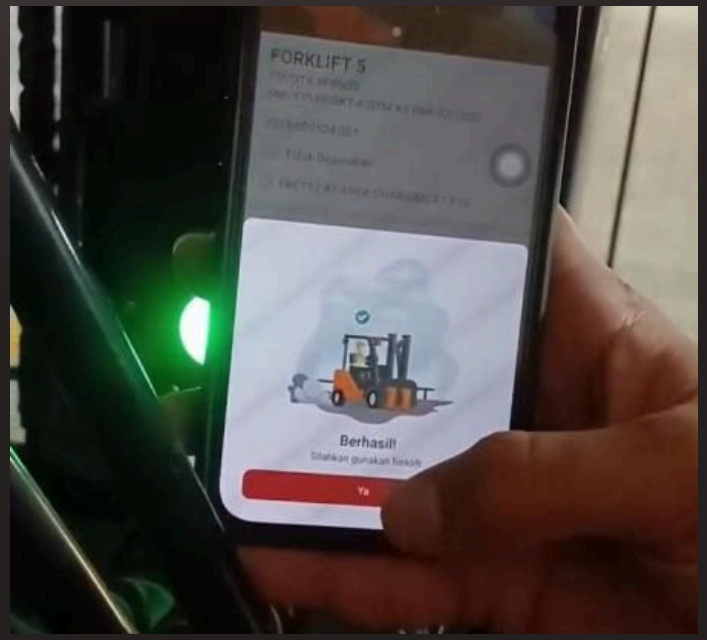
AI Digital SOP

Real-time assembly validation against digital SOPs — OpenCV & MediaPipe Hands, Arduino LED/buzzer feedback, cycle-time analytics. PT Sat Nusapersada · Feb 2026



IoT Air Quality Monitoring System

ESP32 with MQ-7 CO and SEN0322 oxygen sensors, MQTT, FastAPI, and a real-time dashboard with warnings, logging and trends. Freelance · Sep 2026



IoT Forklift Access Control & Fleet Telemetry

Arduino Mega 2560 + RFID ignition interlock, RTL8720DN Wi-Fi, cloud credential validation, and operator logging. PT Sat Nusapersada · Feb–Mar 2026



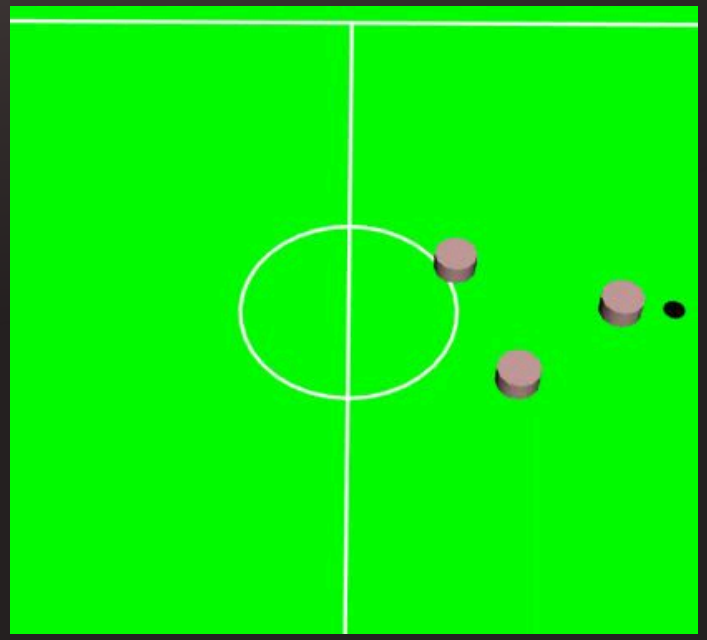
AGV Robot Sumitomo

Automated guided vehicle work — robot wiring, the HMI–PLC communication protocol, PLC programming, sensor setup and debugging. Polibatam · Oct–Dec 2025



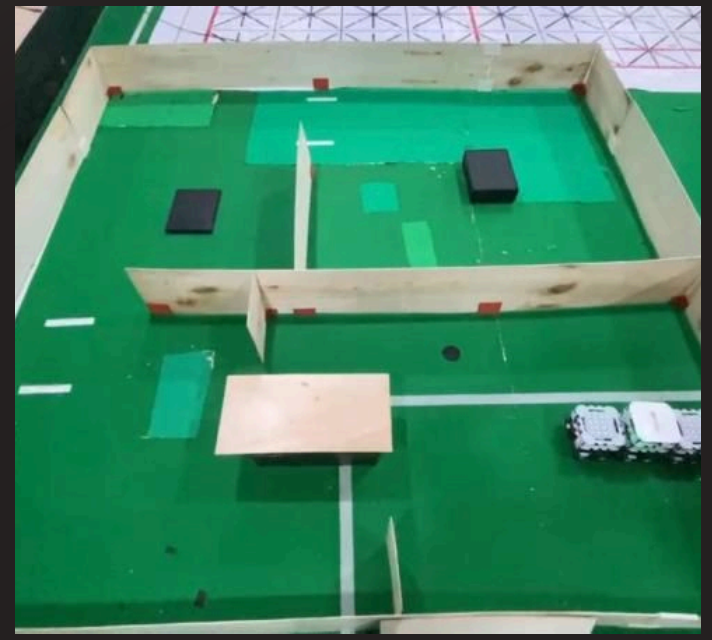
Robot Soccer MSL Research

Research & programming-division lead at BRAIL — strategy, UDP communication, navigation and planning, kinematics, ROS / ROS 2. Polibatam · 2022–2025



Potential Field with Escape Force

A local-minima solution for wheeled soccer robot navigation, simulated and implemented in C++ with Qt Creator. Published at ICAE 2025.



Smorphi Robot — NRC Singapore

Maze-cleaning autonomous navigation and teleoperation with ROS and a monitoring UI. 2nd & 3rd place, Autonomous and Pilot categories, 2024 & 2025.



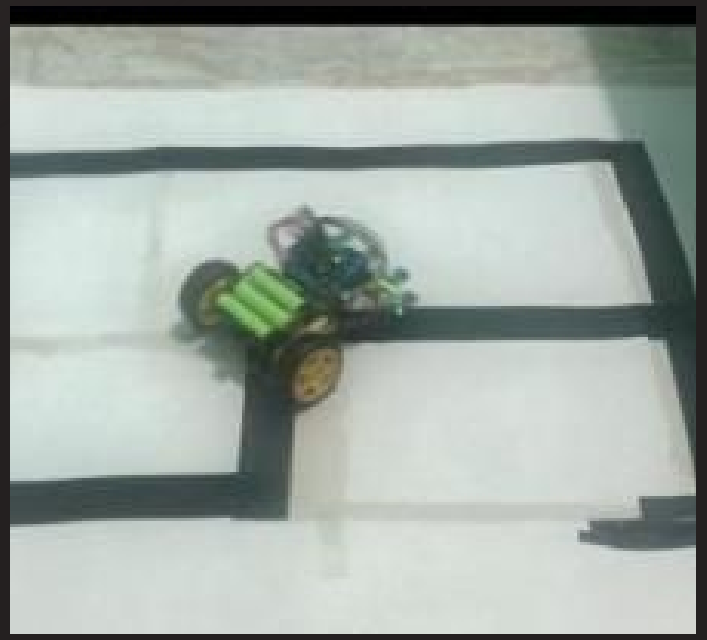
EPSON Industrial Robot — NIRC 2025

EPSON robot system for the National Industrial Robot Competition 2025 — sensor wiring, EPSON RC+ motion programming, palletizing, calibration.



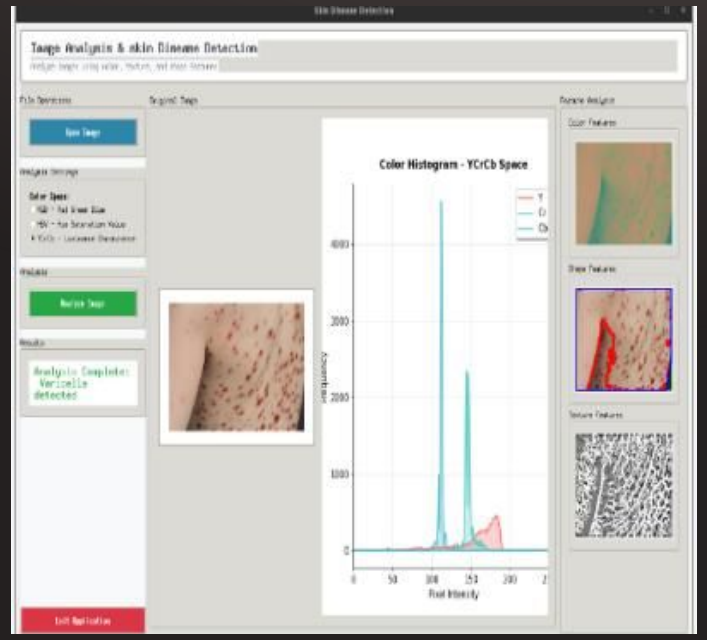
Micromouse Robot

Maze solving in C++ with FloodFill and backtracking algorithms — developed in simulation, validated on the real robot. Freelance · Sep–Nov 2025



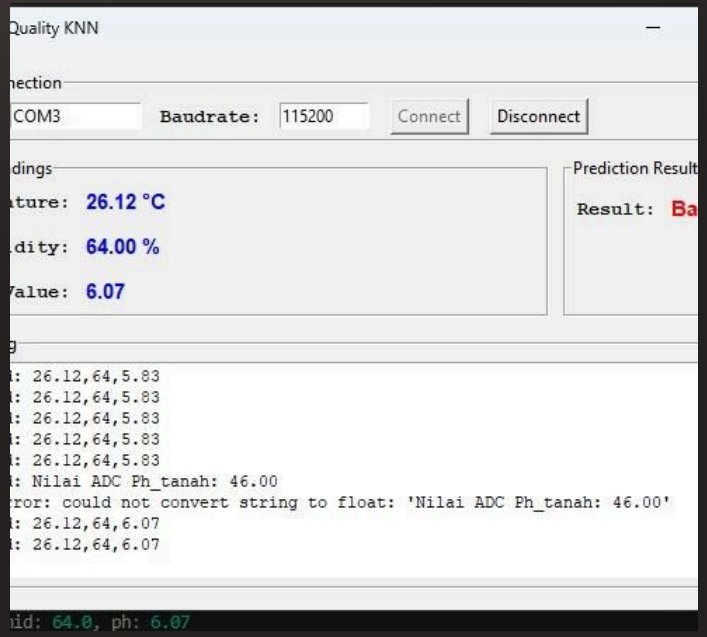
ANN Line Follower Robot

An Arduino-based line follower whose control policy is an artificial neural network, trained and tuned on the physical robot. Freelance · Nov–Dec 2025



ML KNN — Skin Disease Prediction

Image feature extraction and engineering feeding a K-Nearest Neighbors classifier, wrapped in a prediction interface. Oct–Nov 2025



ML KNN — Compost Quality

Exploratory data analysis and feature engineering for compost quality prediction with KNN, deployed on ESP32 with a monitoring UI. Freelance · 2025



Trainer Kit Microcontrollers & IoT

Microcontroller & IoT trainer kit — embedded development, user documentation, learning materials, Android app design. Wirausaha Merdeka · 2024





PKM Karsa Cipta — Plate Detection & OCR

HALCON deep-learning dataset work and Jetson Nano deployment for license plate detection and OCR.

Favorite Champion, PKM-Karsa Cipta at the 37th PIMNAS · Jul–Oct 2024

PUBLICATIONS

Ball Distance Estimation Using Machine Learning on an Omnidirectional Camera for a Middle Size Soccer Robot
ICODSE 2025 · Jan 2026

Improved Navigation of Wheeled Soccer Robots Using Artificial Potential Field with Escape Force
ICAE 2025 · Dec 2025

Implementation of Intercept Ball Algorithm for Wheeled Soccer Robot Barelang63
ICAE · Oct 2024

Implementation of RRT for Path Planning and Stanley Control Method
ICAE · Oct 2023

SKILLS

Python · C++ · C# · JavaScript — ROS / ROS 2 · FastAPI · PostgreSQL — ESP32 · Arduino · STM32 · Raspberry Pi · NVIDIA Jetson — PLC · EPSON RC+ · Modbus · MQTT · YOLO · OpenCV · OCR

HONORS

2nd Place — Smorphi Autonomous Challenge · NRC Singapore 2025

3rd Place — Smorphi Pilot Challenge · NRC Singapore 2025

Favorite Champion — PKM-Karsa Cipta · 37th PIMNAS, 2024

1st Place — Indonesian Wheeled Soccer Robot Contest, Regional · KRI 2024

2nd Place — Indonesian Wheeled Soccer Robot Contest, Regional · KRI 2023

2nd Place — Smorphi Pilot Challenge · NRC Singapore 2024

3rd Place — Smorphi Autonomous Challenge · NRC Singapore 2024

1st Place — Robotics and AI Category · Polibatam PBL Expo, 2024

Participant — National Industrial Robot Competition · NIRC 2025

3rd Place — Manufacturing Fabrication Process · PBL Expo Workshop, 2023

THANK YOU

That's my portfolio so far.

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https://masdikaaliman.github.io/